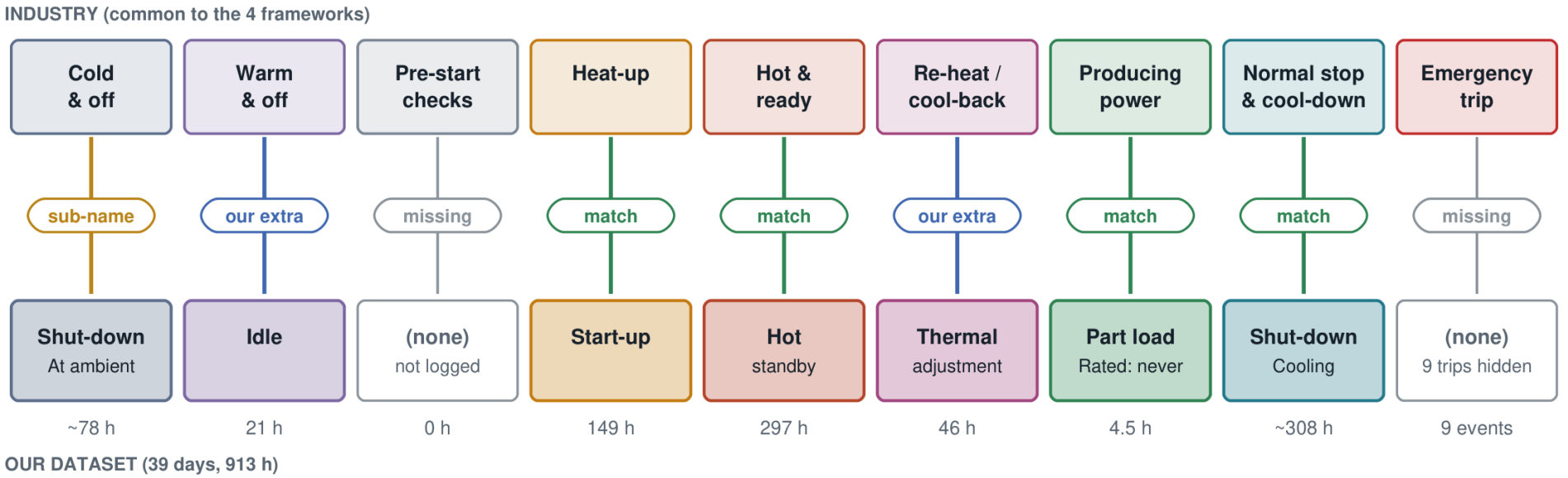
Our stages against the world's

Our six stages follow the same loop that fuel-cell standards, SOFC suppliers, factory automation and power plants all use. Four line up one-to-one, one is filed as a sub-name, two are our own extra detail, and two that industry treats as stages are missing: pre-start checks and the emergency trip.

Why it matters: shared names make our data readable without a translation sheet, and the gaps show what the fake-data generator cannot make yet (there is no trip stage to copy from).



Top: the nine stage families the four frameworks share. Bottom: what our dataset calls the same minutes, with hours.

Contents: 1 Overview · 2 How the world names stages · 3 Side by side · 4 Our stage map · Sources

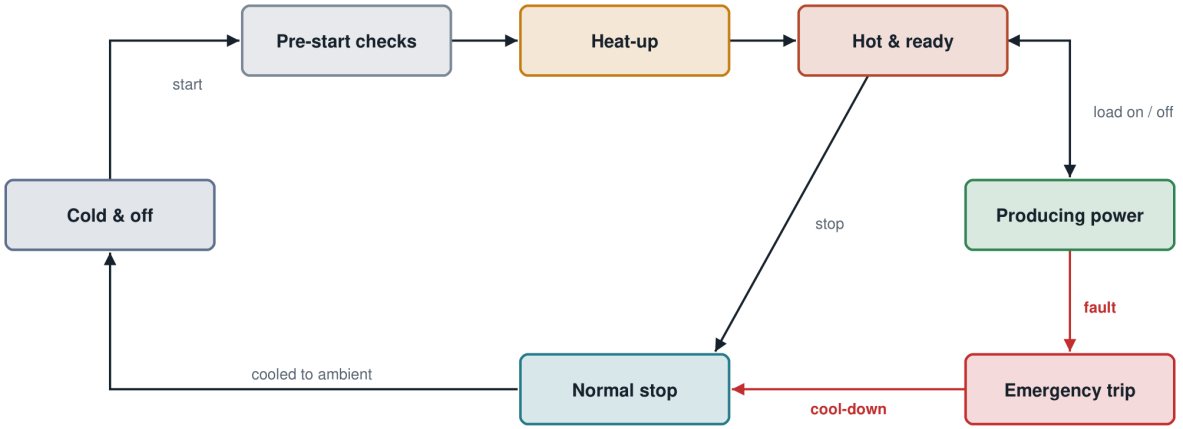
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If you remember one thing

Our stages fit the world's usual cycle. The two gaps are pre-start checks, which our log cannot see, and trips, which happen 9 times but have no stage of their own.

How the world names stages

Four widely used frameworks name the stages of a machine's day. The words differ, but all share the same loop.



The shared loop. A trip is a separate exit that skips the calm ramp-down.

IEC 62282-3-200
International test standard for stationary fuel cells
1. Cold state: ambient, no power in or out
2. Counted inside start-up time
3. Start-up: cold state to net power
4. Pre-generation state: hot, zero power
5. Power generation: rated, partial, minimum
6. Shutdown: back to cold state
7. Emergency shutdown (safety part 3-100)

SOFC plant practice
Reference ladder S0-S8, supplier patents, durability studies
1. S0 Cold standby, anode sealed under N2
2. S1 Pre-start checks: leak test, purge
3. S2 Heat-up under protective gas, S3 reduction and fuel-in
4. S6 Hot standby, 400-600 °C, trickle fuel
5. Inside heat-up or standby
6. S4 Initial load, S5 Normal operation
7. S7 Controlled shutdown, fuel to N2
8. S8 Emergency shutdown (trip)

PackML / ISA-88
Machine and batch automation state model
1. Stopped
2. Stopped
3. Resetting
4. Starting
5. Idle / Held
6. Holding / Unholding
7. Execute
8. Stopping, then Stopped
9. Aborting, Aborted, Clearing

Power plants
Gas and steam turbine practice
1. Offline, cold
2. Offline, warm (sets a warm start)
3. Pre-start checks, purge
4. Cold / warm / hot start
5. Hot standby, synchronised at no load
6. Online, load following
7. Normal shutdown, cool-down
8. Unit trip

Start classes, borrowed from power plants: hot after less than about 10 h offline, warm after 10-72 h, cold after more than 72 h; many plants now use metal temperature instead of the clock.

If you remember one thing

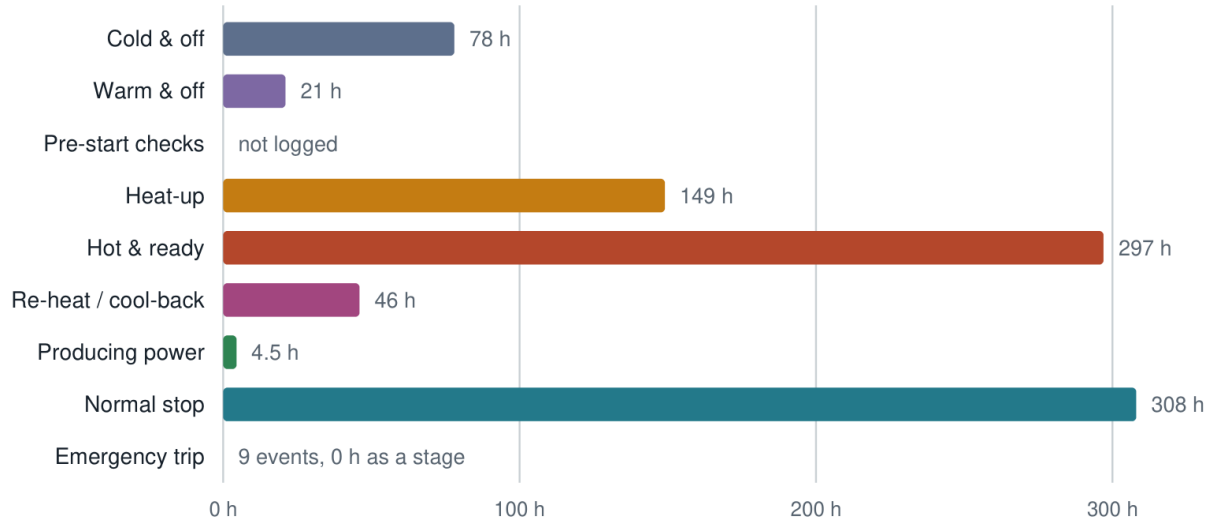
Whatever the words, every framework runs cold, get ready, heat, hot and ready, produce, stop, with a separate exit for faults.

Side by side

Our data covers seven of the nine families. The differences are filing, detail and two real gaps.

Framework	Cold & off	Warm & off	Pre-start checks	Heat-up	Hot & ready	Re-heat / cool-back while hot	Producing power	Normal stop & cool-down	Emergency trip
IEC 62282-3-200	■ Cold state: ambient, no power in or out	■ No name	■ Counted inside start-up time	■ Start-up: cold state to net power	■ Pre-generation state: hot, zero power	■ No name	■ Power generation: rated, partial, minimum	■ Shutdown: back to cold state	■ Emergency shutdown (safety part 3-100)
SOFC plant practice	■ S0 Cold standby, anode sealed under N2	■ No name (counts as cooling)	■ S1 Pre-start checks: leak test, purge	■ S2 Heat-up under protective gas, S3 reduction and fuel-in	■ S6 Hot standby, 400-600 °C, trickle fuel	■ Inside heat-up or standby	■ S4 Initial load, S5 Normal operation	■ S7 Controlled shutdown, fuel to N2	■ S8 Emergency shutdown (trip)
PackML / ISA-88	■ Stopped	■ Stopped	■ Resetting	■ Starting	■ Idle / Held	■ Holding / Unholding	■ Execute	■ Stopping, then Stopped	■ Aborting, Aborted, Clearing
Power plants	■ Offline, cold	■ Offline, warm (sets a warm start)	■ Pre-start checks, purge	■ Cold / warm / hot start	■ Hot standby, synchronised at no load	■ No name	■ Online, load following	■ Normal shutdown, cool-down	■ Unit trip
Our dataset	■ Shut-down, sub-name At ambient	■ Idle	■ Not in data	■ Start-up	■ Hot standby	■ Thermal adjustment	■ Part load (Rated power defined, never reached)	■ Shut-down, sub-name Cooling	■ Not a stage: 9 trips sit at the end of other stages

■ match ■ sub-name / partial ■ our extra ■ missing or no name



Where our 913 hours went, to scale. Producing power is 4.5 h; the two missing families have no hours.

Six differences

1. **Cold & off is its own stage everywhere else.** We file it as a sub-name of Shut-down (At ambient, about 78 h). Same minutes, different shelf.

2. **Pre-start checks are missing.** Industry has a check step before heat-up (leak test, purge done). Our log has no leak-test or purge-done signal, so the short air purge sits inside Start-up.

3. **Re-heat / cool-back is our own extra stage.** Others hide it inside heat-up or standby. We keep Thermal adjustment (18 blocks, 46 h) because its temperature moves while the stack is hot and empty.

4. **Trips are not a stage.** Every framework has a separate fault exit. In our data 9 blocks end in a trip (8 of the 10 Part load blocks), but the trip minutes are labelled Part load, then Shut-down.

5. **Rated power is only on paper.** Rated is 30 A per string (50 kW AC). The highest we saw was 16.2 A and 33.6 kW DC, so all load is Part load.

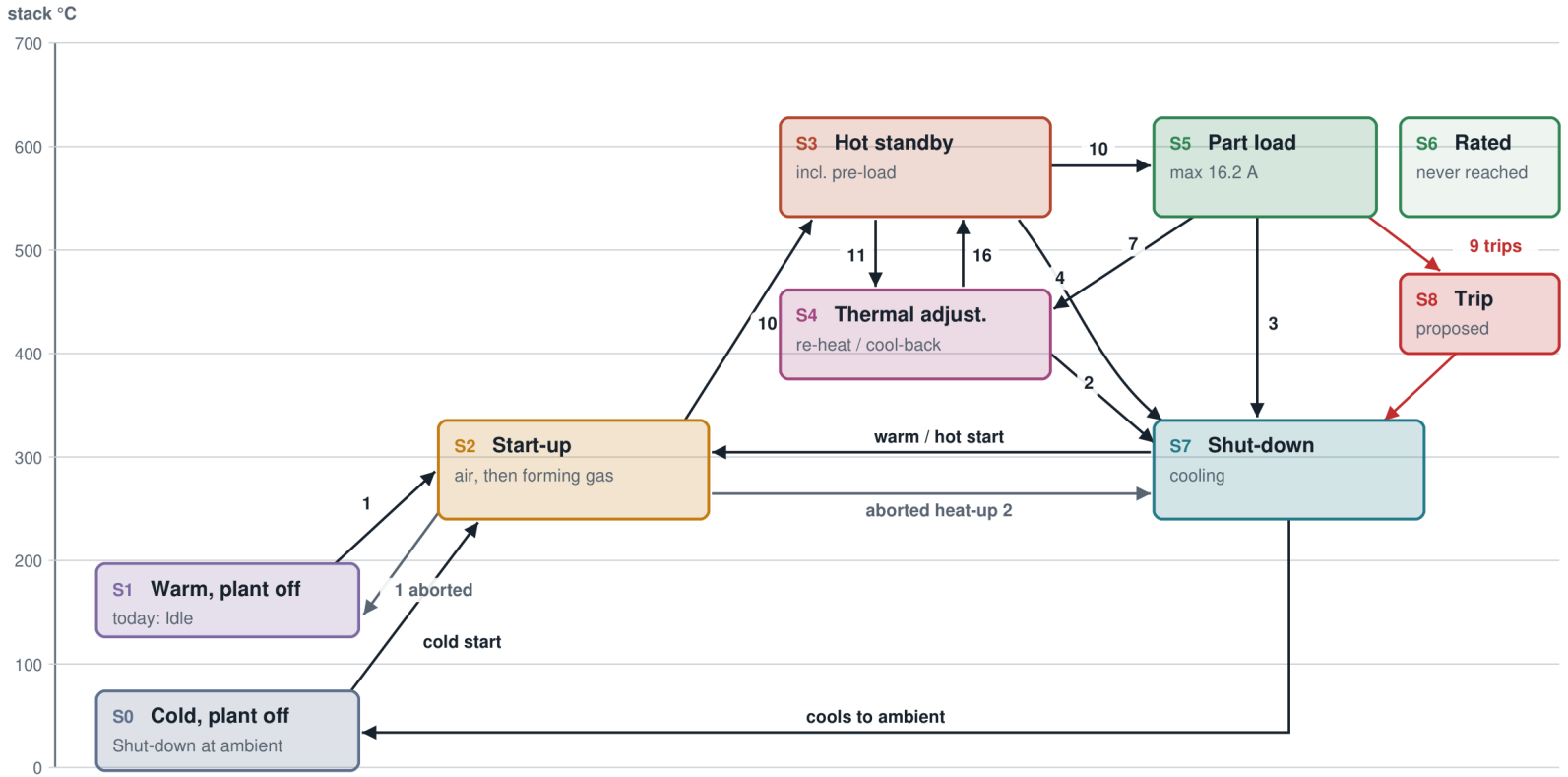
6. **Starts are not classed.** Power plants call every start cold, warm or hot. Our 14 Start-ups split 7 cold (below 60 °C), 4 warm (130-280 °C) and 3 hot (above 450 °C).

If you remember one thing

We are not missing time, we are missing names: the trip and check minutes exist but are filed under other stages.

Our stage map

Our data drawn as nine stages, S0 to S8, in the order and with the names industry uses. Seven are already in our labels, Rated power is never reached, and Trip is the one new stage worth adding.



Boxes sit at the stack temperature where the stage lives. Numbers on arrows = how often that change happened. Red dashed = trip path; grey = outside the rules.

The stage list

Code	Stage (today)	Industry name	How the data shows it	Blocks	Hours
S0	**Cold, plant off** today: Shut-down · At ambient	Cold state / Stopped / Offline cold	No current, heater off, air off, stack within 30 K of ambient	8 pieces	~78
S1	**Warm, plant off** today: Idle	Offline warm (power plants only)	Plant off, stack more than 30 K above ambient, no shut-down before it	2	21
S2	**Start-up** today: Start-up	Start-up / Starting / Heat-up	Heater on with air, stack rising more than 10 °C/h. Air only first; forming gas (N2 + H2) on at 190-270 °C. Class: cold 7, warm 4, hot 3	14	149
S3	**Hot standby** today: Hot standby	Pre-generation / Hot standby / Idle	No load, stack temperature steady (within 5 °C/h), 550 °C or more, or 450-550 °C held 6 h. Includes fuel switch-over and pre-load (natural gas on 5-20 min before load)	26	297
S4	**Thermal adjustment** today: Thermal adjustment	No standard name (PackML Holding is closest)	Hot, no load, temperature moving 5 °C/h or more (re-heat or cool-back)	18	46
S5	**Part load** today: Part load	Power generation, partial / Execute	String current 1-30 A, always on natural gas	10	4.5
S6	**Rated power** today: Rated power	Power generation, rated	About 30 A per string. Never reached	0	0
S7	**Shut-down** today: Shut-down · Cooling	Shutdown / Stopping / Normal shutdown	Load off, stack falling. Forming gas kept down to 210-290 °C, then air only, then natural cooling with air off	14 pieces	~308
S8	**Trip (proposed)** today: Not a stage	Emergency shutdown / Aborted / Unit trip	A protective trip (most often exhaust TIRZ 8,1,10 at 200 °C or more) until the cool-down is under way	9 events	0

Your decisions

	Choice	Recommendation
A	**Show the S0-S8 codes and industry names next to today's names.**	Display only. No rule or file changes. Recommended.
B	**Trip: own stage S8, or a flag column?**	Recommend a flag column first: only 9 events, but the fake-data generator needs it to add trips.
C	**Add a start class (cold / warm / hot) to every Start-up.**	Zero risk; helps match the pattern filler by start temperature. Recommended.
D	**Make At ambient its own stage S0 (as industry does).**	Changes stage folders, generator and checker. Recommend keeping it a sub-name and showing S0 only as a label.

If you remember one thing

Codes S0-S8 give each of our stages its industry name without changing a single rule; the only real addition is a Trip stage or flag.

Sources

IEC 62282-3-200 (stationary fuel cell performance tests): definitions of cold state and pre-generation state: https://standards.iteh.ai/catalog/standards/iec/81044e7e-8bbd-4a8d-a589-c7e4a92156a3/iec-62282-3-200-2015

IEC 62282-3-200:2025, third edition (revised terms and start-up / shutdown test): https://www.evs.ee/en/evs-en-iec-62282-3-200-2025

Reference SOFC sequence ladder S0-S8 (your earlier page): https://claude.ai/artifact/MJPsTuaZf55stNV4Q6NoJd

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Protective gas on the anode during trips, shut-down and hot standby (patent): https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9005827

Hanasaki et al., J. Electrochem. Soc. 2014: hot standby, cold standby and shutdown as the main interruptions: https://strathprints.strath.ac.uk/85587/1/Hanasaki_etal_JES_2014_SOFC_durability_against_standby_and_shutdown.pdf

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